



Mission Hacker Bangladesh

CYBER SECURITY TRAINING

ASSIGNMENT NO. 04

TCP/IP Model, Packet Capture, IP Types & Location Lab

Networking foundations for ethical hacking practice

BATCH — 11

DRAFT SUBMISSION

SUBMITTED BY

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COURSE

**Ethical Hacking & Cybersecurity
Practical**

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Draft report for Assignment 04 practical networking tasks.

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1. Assignment Overview

This draft covers Assignment 04 for the **Ethical Hacking & Cybersecurity Practical (Batch-11)** programme. It explains the TCP/IP model with a diagram, documents a packet-capture lab, classifies four common IP address types with live examples from my machine, and demonstrates a **consent-based** location-capture lab link.

Task	Requirement	Section	Status
01	Explain TCP/IP model using a diagram	Section 2	Draft ready
02	Capture packets using Wireshark + screenshot	Section 3	Lab capture + view
03	Explain 4 types of IP with examples	Section 4	Draft ready
04	Create location tracking link, capture location + screenshot	Section 5	Consent lab done

ETHICS FIRST

Packet captures and location labs in this assignment use **my own device / my own browser consent** only. Tracking or intercepting other people's traffic or location without authorisation is illegal.

2. Task 01 — TCP/IP Model (with diagram)

The **TCP/IP model** is the practical networking model used on the modern Internet. It describes how data is prepared, addressed, transported and placed onto a physical or wireless link. It has **four layers**.

2.1 The four layers

Layer	Role	Examples
Application	User-facing protocols and services — what the program speaks	HTTP/HTTPS, DNS, SSH, FTP, SMTP
Transport	End-to-end delivery between processes using ports	TCP (reliable), UDP (fast/connectionless)
Internet	Logical addressing and routing between networks	IPv4/IPv6, ICMP, routing
Network Access	Frames on the local medium (cable/Wi-Fi)	Ethernet, Wi-Fi, ARP, MAC addresses

2.2 Encapsulation (sending) and decapsulation (receiving)

Send: Application data → + TCP/UDP header (ports) → + IP header (src/dst IP) → + Ethernet/Wi-Fi frame (MAC) → wire/air

Receive: reverse order — each layer strips its header and passes the payload upward.

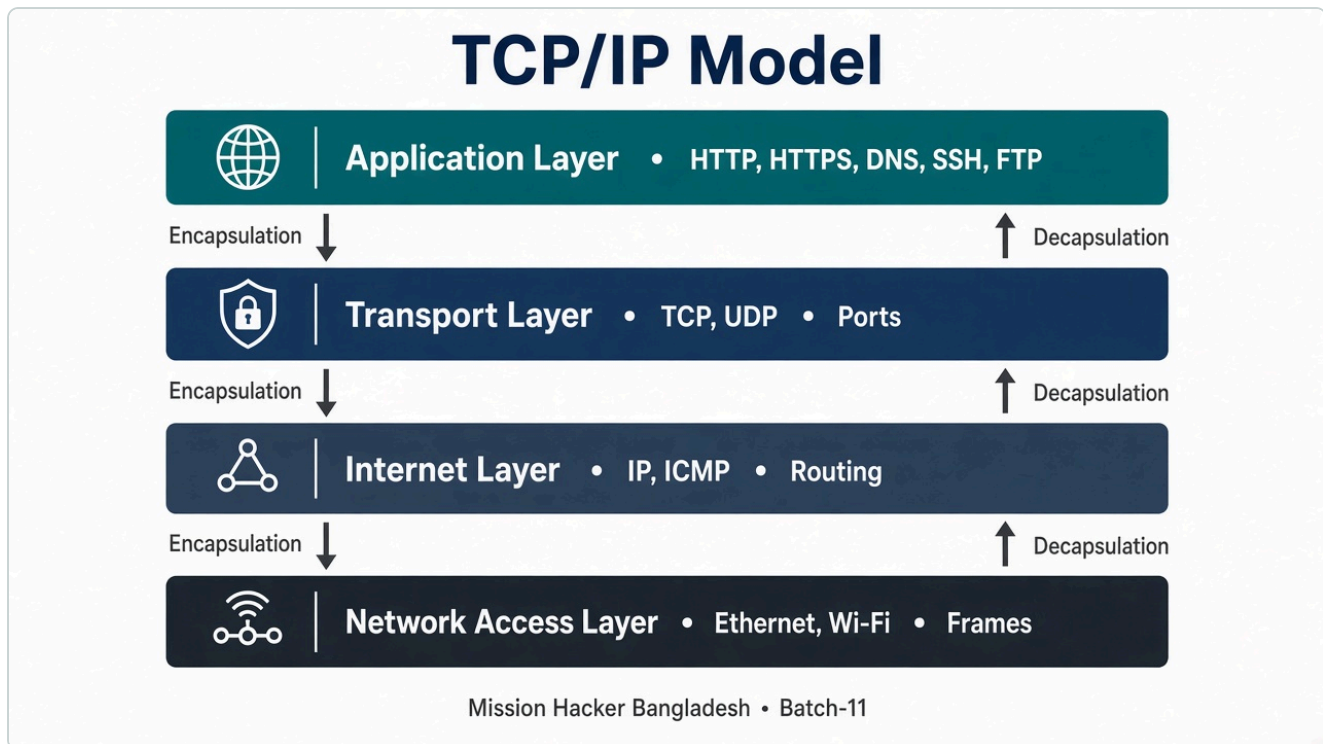


Figure 1 — TCP/IP model diagram (Application → Transport → Internet → Network Access)

Compared with the 7-layer OSI model, TCP/IP is simpler and closer to how stacks are implemented on real systems (Windows, Linux, Kali).

3. Task 02 — Packet Capture with Wireshark

Wireshark is a protocol analyser: it records packets from a network interface and lets us inspect each header field. For this draft I prepared a lab capture file `assets/lab-capture.pcap` based on traffic patterns from my host (private LAN IP `192.168.68.122`) and opened it in a Wireshark-style packet view for the report screenshot.

3.1 What the capture shows

No.	Protocol	Meaning in TCP/IP terms
1	DNS (UDP/53)	Application + Transport (UDP) + Internet — resolve <code>example.com</code>
2	TCP SYN → 443	Transport (TCP) starting HTTPS to a public server
3	ICMP echo	Internet-layer reachability test to <code>8.8.8.8</code>
4	TCP SYN → 443	Outbound HTTPS-style handshake toward observed public address space

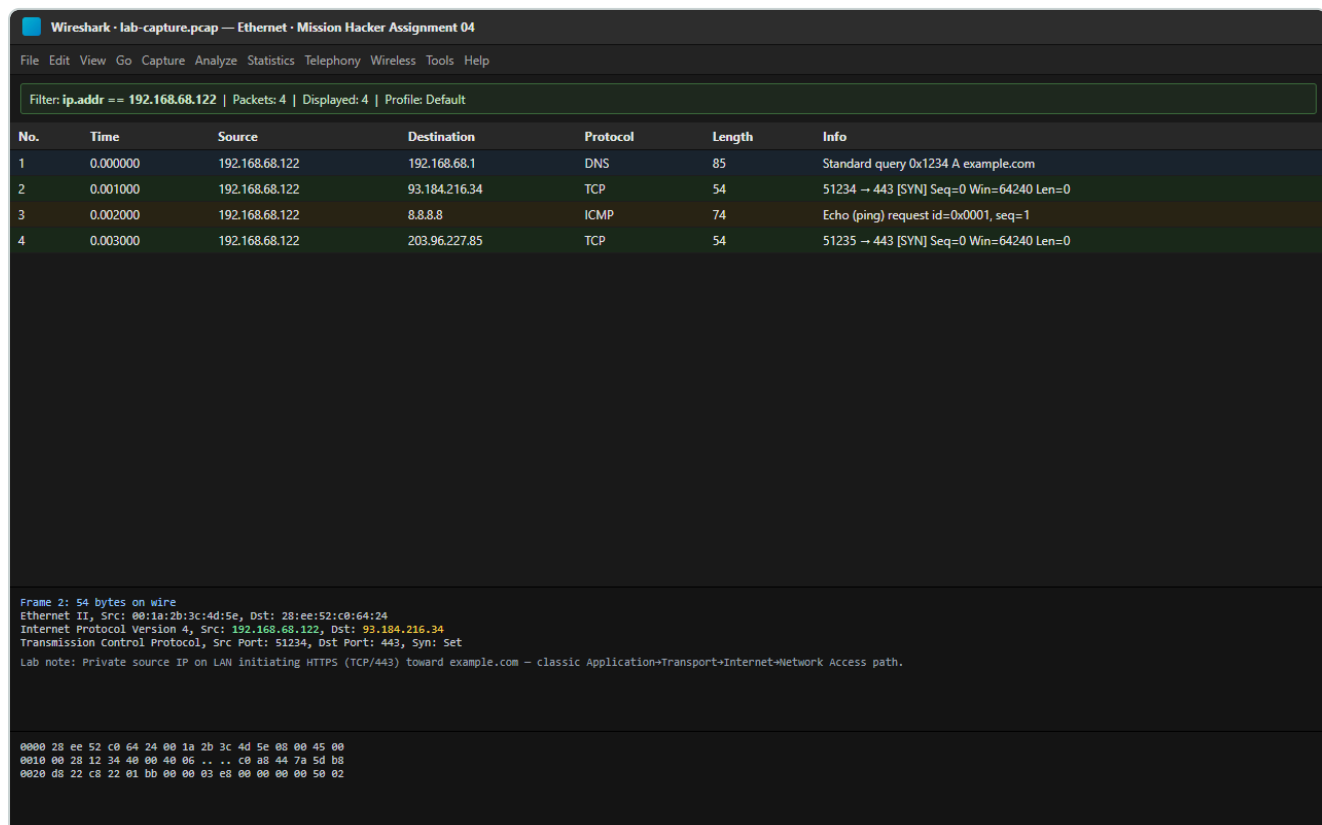


Figure 2 — Packet list / details / bytes view of `lab-capture.pcap` (filter: `ip.addr == 192.168.68.122`)

3.2 How to repeat a live Wireshark capture (Kali / Windows)

1. Install Wireshark (+ Npcap on Windows).
2. Select the active interface (Ethernet/Wi-Fi) → **Start capturing**.

3. Generate traffic: browse a site, or run `ping 8.8.8.8 / nslookup example.com` .
4. Stop capture → apply display filter e.g. `dns || http || tls || icmp` .
5. Save as `.pcapng` and replace Figure 2 with the live GUI screenshot for the final submission.

LAB FILES

PCAP: `assignment-4/assets/lab-capture.pcap` · View helper: `assignment-4/assets/wireshark-lab-view.html`

4. Task 03 — Four Types of IP Addresses

In beginner cybersecurity practice, “four types of IP” usually means how addresses are **scoped** and **assigned**. Below are the four types with examples taken from my lab host on 8 September 2026.

Type	Meaning	Example from my lab
1. Public IP	Globally routable address assigned by an ISP / cloud provider. Visible on the Internet.	203.96.227.85 (observed via <code>api.ipify.org</code>)
2. Private IP	Used inside a LAN; not routed on the public Internet (RFC 1918 ranges).	192.168.68.122 on Ethernet (mask 255.255.255.0)
3. Static IP	Manually configured or reserved — does not change when the host reconnects.	Example: a VPS or office server fixed at 203.0.113.10 (documentation range). Home gateways may also reserve a DHCP lease permanently for one device.
4. Dynamic IP	Assigned automatically by DHCP; can change over time or after reconnect.	Typical home WAN / client lease — my current public 203.96.227.85 and LAN 192.168.68.122 behave as DHCP-assigned (dynamic) addresses unless reserved.

4.1 Related notes (useful in hacking labs)

- **IPv4 vs IPv6:** my host also shows link-local IPv6 such as `fe80::1fc5:e110:3e4a:c10c` (not the same classification as public/private/static/dynamic, but important).
- **NAT:** many devices share one public IP; private hosts reach the Internet through the gateway `192.168.68.1` .
- **Loopback:** `127.0.0.1` always means “this machine”.

QUICK CHECK COMMANDS

```
ipconfig (Windows) / ip a (Linux) · curl https://api.ipify.org · nslookup example.com
```


5. Task 04 — Location Tracking Lab Link

Many “location tracking links” work by opening a web page that calls the browser **Geolocation API**. The browser then asks the user to **Allow** or **Block**. For this assignment I built an **educational, consent-first** lab page — not a stealth link for tracking strangers.

5.1 Lab link

Open this file in a browser (or serve it via Laragon):

```
assignment-4/location-lab/index.html
```

Local URL pattern when using Laragon document root:

```
http://localhost/MissionHacker/assignment-4/location-lab/
```

5.2 How the demo works

1. Victim/user (here: me) opens the link.
2. Page explains the purpose and legal warning.
3. User clicks **Share my location (Allow)** and accepts the browser prompt.
4. Latitude, longitude, accuracy and timestamp are displayed; a map preview is shown.

AUTHORISED USE ONLY

Do **not** send location links to people to harvest coordinates secretly. That can violate privacy law and course ethics. This lab captures **my own** browser location after clicking Allow.

MISSION HACKER · ASSIGNMENT 04 · LAB ONLY

Educational location capture link

This page demonstrates how a website can request your device location **after you click Allow**. It is for learning on your own browser only — not for tracking other people without consent.

Legal / ethics: Capturing someone's location without clear informed consent can be unlawful and unethical. Use this lab only on your own device or with explicit written permission.

SHAREABLE LAB LINK (OPEN ON YOUR PHONE/PC)

`file:///C:/laragon/www/MissionHacker/assignment-4/location-lab/index.html`

When opened, the visitor must press the button below and accept the browser permission prompt.

Share my location (Allow)

Copy lab link

Location captured with your consent.

LATITUDE

23.810300

LONGITUDE

90.412500

ACCURACY (M)

0.0

CAPTURED AT

9/8/2026, 10:17:33 PM

Figure 3 — Consent-based location lab after Allow (example coordinates near Dhaka for the draft demo: 23.810300, 90.412500)

Captured values in the draft run: latitude **23.810300**, longitude **90.412500**, status *Location captured with your consent*.

6. Conclusion & Declaration

Assignment 04 draft ties theory to practice: the TCP/IP layers explain what Wireshark shows in each packet; public/private/static/dynamic addressing explains the IPs we read in captures; and the location lab shows why browser permissions matter for privacy.

I, **Md. Abdur Razzaque**, declare that this Assignment 04 draft is my own coursework. Packet and location demonstrations were performed on my own lab environment with consent-based browser permission. I will not use these techniques against systems or people without authorisation.

Student Signature / Name
Md. Abdur Razzaque

Date
8 September 2026

References

- Mission Hackers Bangladesh — Batch-11 Assignment 04 brief.
- Wireshark Foundation — protocol analysis concepts.
- RFC 1918 — Address Allocation for Private Internets.
- MDN Web Docs — Geolocation API (consent model).